

① mapeamento direto

End. dec.	End. binário	Tag	Índice	Offset	Hit/Miss
0	<u>00000000</u> ₂	000	000	00	miss
4	00000100 ₂	000	001	00	miss
8	00001000 ₂	000	010	00	miss
0	00000000 ₂	000	000	00	hit
32	00100000 ₂	001	000	00	miss
4	00000100 ₂	000	001	00	hit
36	00100100 ₂	001	001	00	miss
8	00001000 ₂	000	010	00	hit
40	00101000 ₂	001	010	00	miss
0	00000000 ₂	000	000	00	miss
32	00100000 ₂	001	000	00	miss
44	00101100 ₂	001	011	00	miss

$$\text{linhas} = 32/4 = 8$$

$$\text{offset} = \log_2 4 = 2$$

$$\text{índice} = \log_2 8 = 3$$

$$\text{tag} = 8 - 3 - 2 = 3$$

End. dec.	End. binário	Tag	Índice	Offset	Hit/Miss
16	<u>00010000</u> ₂	00	010	000	miss
20	00010100 ₂	00	010	100	hit
24	00011000 ₂	00	011	000	miss
28	<u>00011100</u> ₂	00	011	100	hit
80	01010000 ₂	01	010	000	miss
84	01010100 ₂	01	010	100	hit
16	00010000 ₂	00	010	000	miss
20	00010100 ₂	00	010	100	hit
88	01011000 ₂	01	011	000	miss
92	01011100 ₂	01	011	100	hit
24	00011000 ₂	00	011	000	miss
28	00011100 ₂	00	011	100	hit

offset: $\log_2 8 = 3$ $n^\circ \text{ linhas} = 64/8 = 8$ índice: $\log_2 8 = 3$ tag: $8 - 3 - 3 = 2$

② Associação por conjuntos

End. dec.	End. binário	Tag	Índice	Offset	Hit/Miss
0	<u>00000000</u> ₂	0000	00	00	miss
16	00010000 ₂	0001	00	00	miss
0	00000000 ₂	0000	00	00	hit
4	00000100 ₂	0000	01	00	miss
20	00010100 ₂	0001	01	00	miss
4	00000100 ₂	0000	01	00	hit
32	00100000 ₂	0010	00	00	miss
16	00010000 ₂	0001	00	00	miss
36	00100100 ₂	0010	01	00	miss
20	00010100 ₂	0001	01	00	miss
0	00000000 ₂	0000	00	00	miss
4	00000100 ₂	0000	01	00	miss

offset: $\log_2 4 = 2$ conjuntos: $8/2 = 4$ tag: $8 - 3 - 2 = 3$
linhas: $32/4 = 8$ índice: $\log_2 4 = 2$

End. dec.	End. binário	Tag	Índice	Offset	Hit/Miss
4	<u>00000100</u> ₂	0000	01	00	miss
20	00010100 ₂	0001	01	00	miss
36	00100100 ₂	0010	01	00	miss
52	00110100 ₂	0011	01	00	miss
4	00000100 ₂	0000	01	00	hit
20	00010100 ₂	0001	01	00	hit
68	01000100 ₂	0100	01	00	miss
84	01010100 ₂	0101	01	00	miss
36	00100100 ₂	0010	01	00	miss
52	00110100 ₂	0011	01	00	miss
100	01100100 ₂	0110	01	00	miss
4	00000100 ₂	0000	01	00	miss

offset = $\log_2 4 = 2$ conjunto = $16/4 = 4$ tag = $8 - 2 - 2 = 4$

linha = $64/4 = 16$ índice = $\log_2 4 = 2$

③ completamente associativo

End. dec.	End. binário	Tag	Índice	Offset	Hit/Miss
0	<u>00000000</u> ₂	000000	-	00	miss
4	00000100 ₂	000001	-	00	miss
8	00001000 ₂	000010	-	00	miss
12	00001100 ₂	000011	-	00	miss
0	00000000 ₂	000000	-	00	hit
4	00000100 ₂	000001	-	00	hit
16	00010000 ₂	000100	-	00	miss
20	00010100 ₂	000101	-	00	miss
8	00001000 ₂	000010	-	00	miss
12	00001100 ₂	000011	-	00	miss
16	00010000 ₂	000100	-	00	hit
0	00000000 ₂	000000	-	00	miss

offset = $\log_2 4 = 2$ tag = $8 - 2 = 6$

End. dec.	End. binário	Tag	Índice	Offset	Hit/Miss
0	<u>00000000</u> ₂	00000	-	000	miss
8	00001000 ₂	00001	-	000	miss
16	00010000 ₂	00010	-	000	miss
24	00011000 ₂	00011	-	000	miss
0	00000000 ₂	00000	-	000	hit
8	00001000 ₂	00001	-	000	hit
32	00100000 ₂	00100	-	000	miss
40	00101000 ₂	00101	-	000	miss
16	00010000 ₂	00010	-	000	miss
24	00011000 ₂	00011	-	000	miss
48	00110000 ₂	00110	-	000	miss
0	00000000 ₂	00000	-	000	miss

$$\text{offset} = \log_2 8 = 3 \quad \text{tag} = 8 - 3 = 5$$